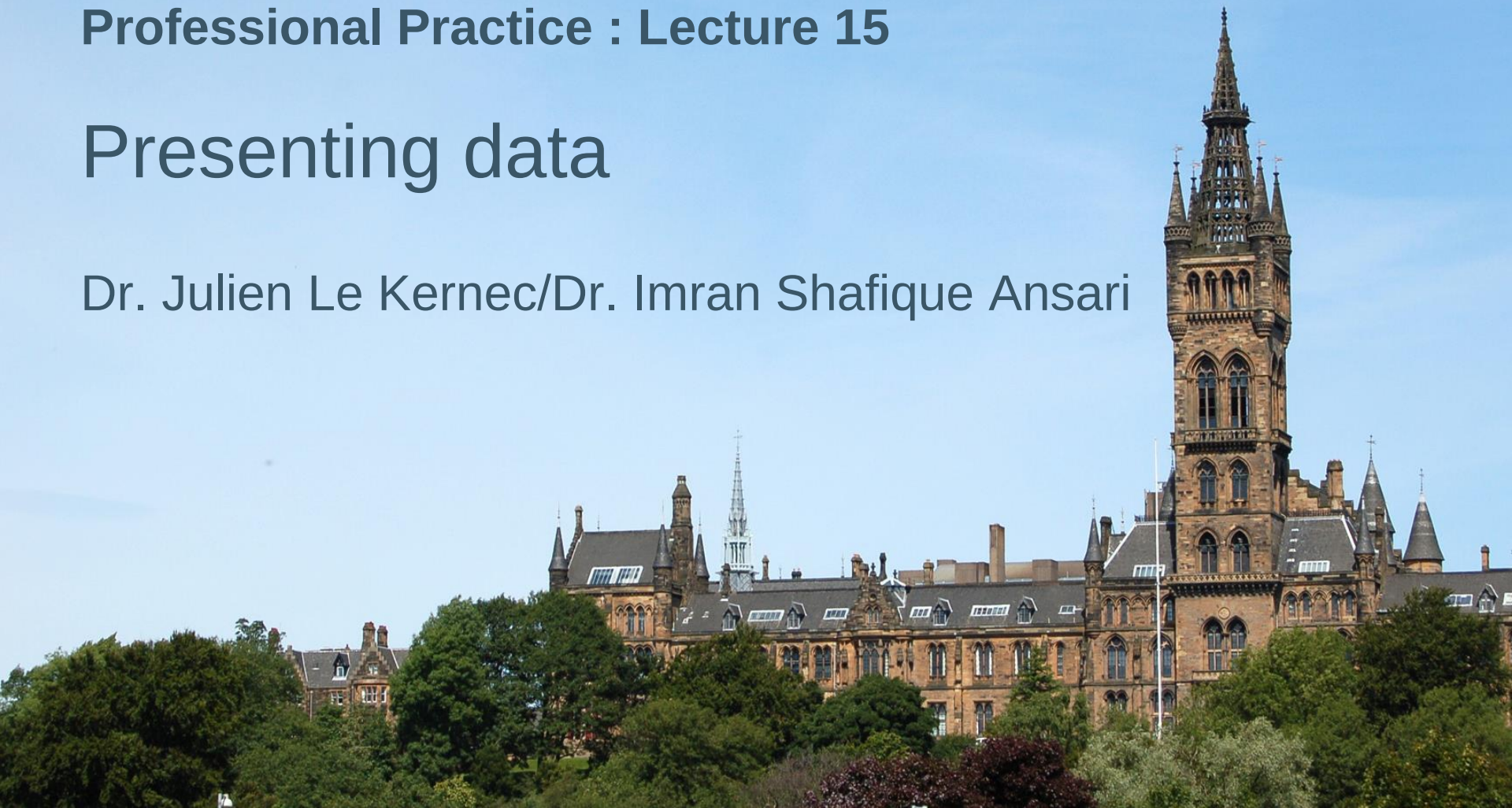
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UoG / UESTC Joint School of Engineering

Professional Practice : Lecture 15

Presenting data

Dr. Julien Le Kernec/Dr. Imran Shafique Ansari

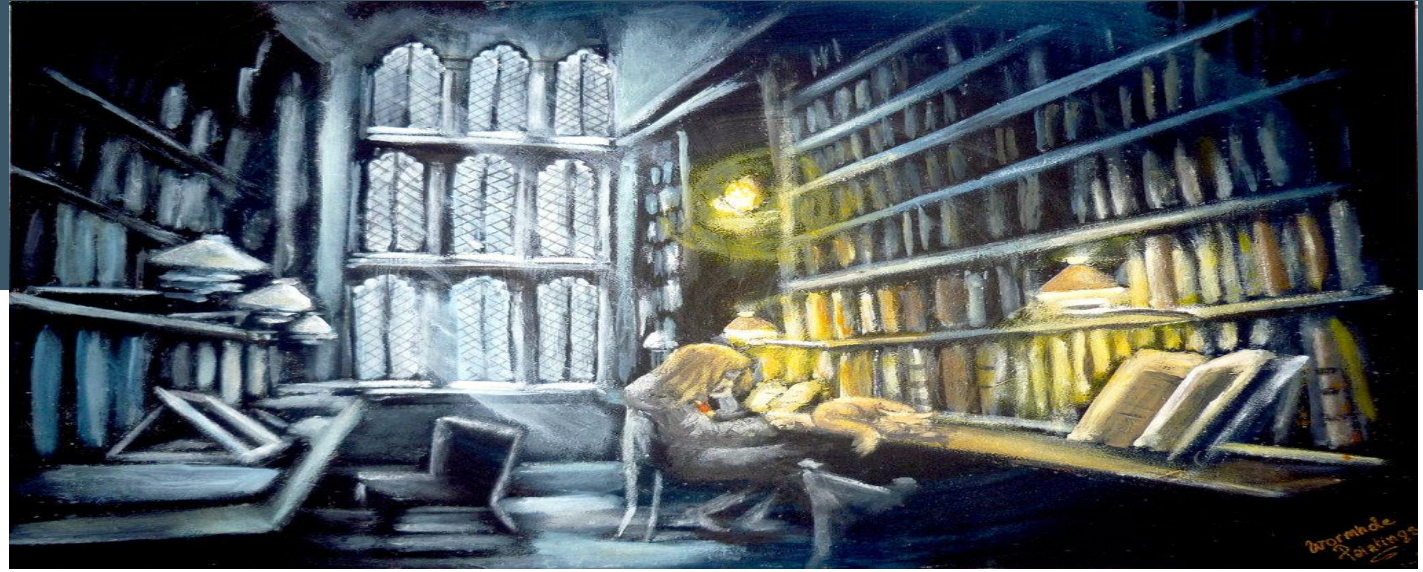


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Outline

- Literature review
- Presenting numerical data
- Infographics
- Writing up a research paper

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Literature Review





A literature review can be a big hurdle if you are not organized

Literature review

- You've seen how to do a literature search
- 1. Before you start a literature search is to have some background information/contextual information about the subject. i.e. do you get the gist of the subject?
- 2. Define a scope for your search: (try it yourself)

Search terms	Bing search	IEEE xplore
Machine learning	8,330,000	72,859
Machine learning in assisted living	71,800	172
Machine learning in assisted living wearables	13,100	28
Machine learning in assisted living wearables or radar	29,100	8

Literature Review

- Keep track of your potential references in a reference manager: Mendeley, Endnote, Bibtex, ….

- A literature review will serve different purposes:
  - What others are doing?
  - Identify a gap in knowledge in a research area?
  - Survey the state-of-the-art and outstanding challenges
  - Background information
  - …



People writing these are usually biased and might not always be honest. So your job is to sort out the relevant information objectively.

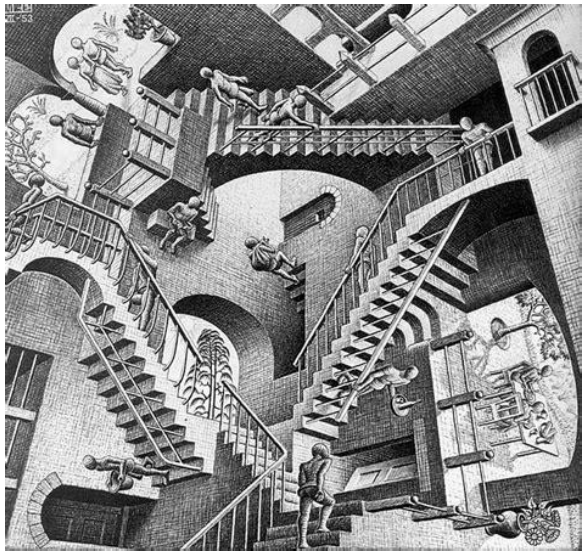
Essentially, what is the purpose of your research piece and how can my literature review bring context and support my research method/data/findings?

Literature review

- Then comes in how do I present that information in a comprehensive manner so that I can tell that story.

- How can you present research data from literature?
  - Ideas??





Essentially how can I tell someone versed in engineering but not in my exact field the essence of what I read?

Presenting research data



Ever tried drinking from a firehose???

Presenting research data

- Example: 6 months work
- 20GB of raw data and 10x more in processed data (if stored)
- Metrics seven excel pages
- Complex experimental setup/complex programme
- ….
- To be condensed in 4-6 pages for a conference, 8-10 pages for a journal paper, 40 pages for your final year project, 1 page for a poster, 15 mn for a presentation, ….

How do I cope???

The University of Nottingham
UNITED KINGDOM - CHINA - MALAYSIA

ETIS

UPMC SORBONNE UNIVERSITÉS

LIP6


J. Le Kernec


O. Romain

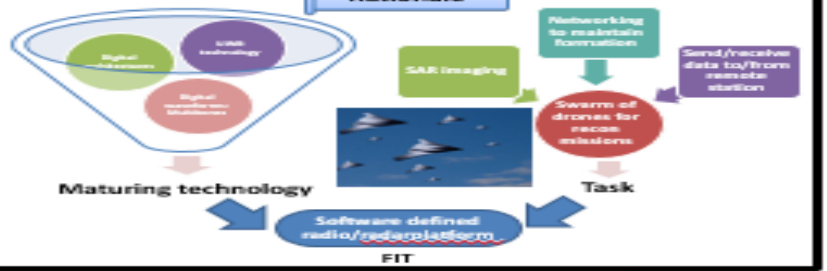

P. Garda


J. Denoulet

Empirical comparison of Chirp and Multitones on experimental UWB Software Defined Radar prototype

Research question: how do Multitones signals compare to Chirp signals in multifunction radar applications?

Rationale



Literature review

Multitones
- Basic radar performances are not well known
- Lack of a viable reference for comparison

Train of diverse multitone pulses
- Near Thumbtack ambiguity function (Levanon et al.,2004)
- No range-Doppler coupling
- Higher pedestal error

New processing capabilities
- Doppler resolution while using agility (van Genderen et al. 2008-1)
- Doppler ambiguity resolution over one pulse train (van Genderen et al.2008-2)

Software-defined radar platforms
- A few lab test beds(ex: (van Genderen et al., 1998)(DRDN/W, 2008)(Pritchard,2007)(Garner,2004)(Time,2006))
- Technology still in its early stage

Software defined platform design



Experimental setup

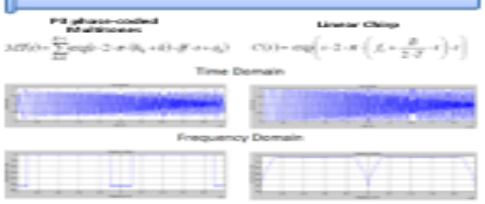

Experimental ground

Comparison to state-of-the-art prototypes

platform	PANOSIA [VanGenderen,98]	SELEX-sens. syst. [Pritchard,07]	LLL imaging and sensors [Garner,04]	[Time,06]	This work
Instantaneous bandwidth	7.5 GHz	800 MHz	500 MHz	20 MHz	800 MHz
Range resolution	0.19 m	0.56	0.3 m	N/A	0.1875 m
Detection range	N/A	140 m	1.5 km/3 km	N/A	200 m
Sampling scheme	Shannon	Sub-Nyquist + Equivalent	Shannon	Shannon	Nyquist
Sampling frequency	N/A	1.95 GS/s	1 GS/s	N/A	2 GS/s
Tuning Range	8 to 12 GHz	10 to 11.5 GHz	7 to 8 GHz	6.4 to 7.5 GHz	10 to 11.6GHz
Architecture	Super-heterodyne + switch-processing	Super-heterodyne + frequency-interleaving	Super-heterodyne	Super-heterodyne + 32 GS/s/ADC + 512 channels	Super-heterodyne
Supported waveforms	Stepped-Multitones, multiband PMCW	Multitones	Multitones	UWB, OOK, IEEE.802.11/a	any
Pulse width	3.125 ns per step	100 to 200 ns	100 to 512 ns	Dependent on standard	7.68 ns up to 1 ms

The Experimental data is recorded on a hard drive and processed offline with MATLAB

Studied Waveforms

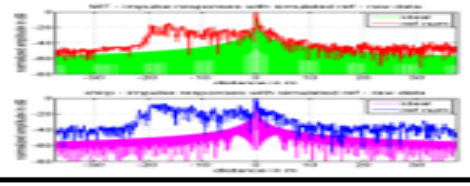


Tested configurations cover most radar applications from surveillance radar to high resolution radar
- Bandwidth-time product > 25
- Tested Bandwidth → 1MHz, 10MHz, 150MHz, 800MHz
- Test Signal period → 500ns, 5µs, 50µs, 500µs, 1ms

Simulations vs Measurements

Parameters' characteristic simulations, relative errors for sidelobe width & sidelobe position and difference in sidelobes amplitudes

Bandwidth	1 MHz	10 MHz	150 MHz	800 MHz
3 dB Mainlobe width	1.8km <1.8 %	180m <1.8 %	0.9m <2.8 %	0.225m <3? %
Sidelobe amplitudes	-13.27dB <0.4 dB	-13.27dB <0.4 dB	-13.27dB <0.4 dB	-13.27dB [~7.4 dB]
Sidelobe positions	±214.3 m <1.7 %	±21.4 m <1.7 %	±1.425 m < 8.1 %	±0.3 m <3? %



Detection range

Chirp yields an extended detection range compared to P3 phase-coded Multitones

simulations 13-17%
measurements 5-19%

Conclusions and perspectives

Software defined radar prototype
- Allows unbiased study of various waveforms
- Performances comparable to state-of-the-art prototypes

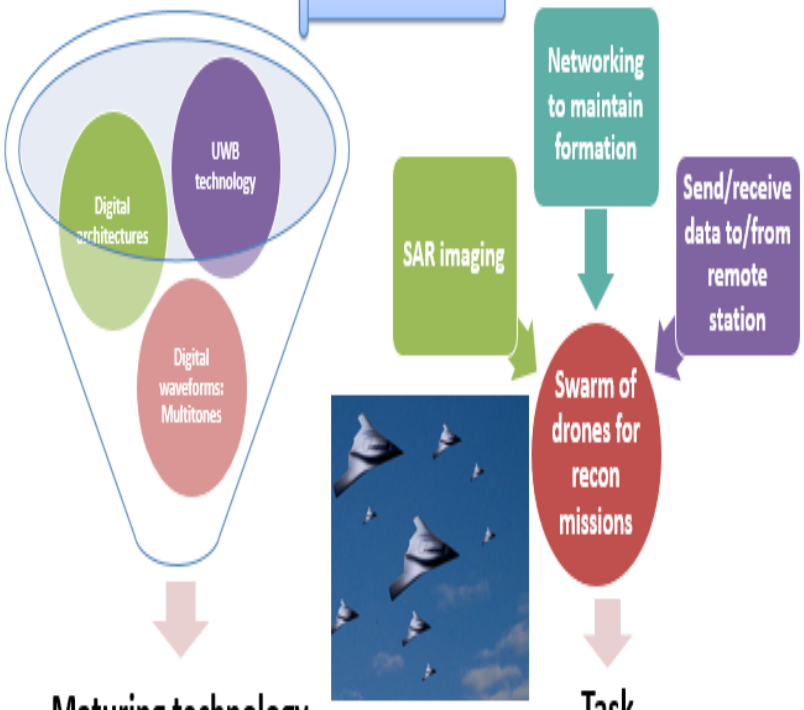
Chirp vs P3 phase coded Multitones
- Same performances in range profile
- 15% extra detection range with Chirp

Future multifunction radar
- Multitones can incorporate telecommunications
- At least 15% loss in detection range for modulation with similar PMEPR reduction capability as P3 codes
- Multitones is the optimal waveform for overall performances (comm+radar)

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Context / Literature Review

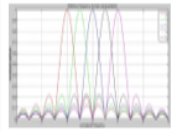
Rationale



Literature review

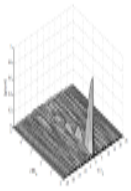
Multitones

- Basic radar performances are not well known
- Lack of a viable reference for comparison



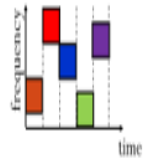
Train of diverse multitone pulses

- Near Thumbtack ambiguity function [Levanon et al.,2004]
- No range-Doppler coupling
- Higher pedestal error



New processing capabilities

- Doppler resolution while using agility [van Genderen et al. 2008-1]
- Doppler ambiguity resolution over one pulse train [van Genderen et al.2008-2]



Software-defined radar platforms

- A few lab test benches [van Genderen et al.,1998][IDROMel, 2009][Paichard,2007][Garmatyuk etal.,2009]
- Technology still in its early stage



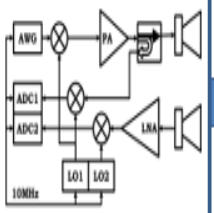
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Methodology

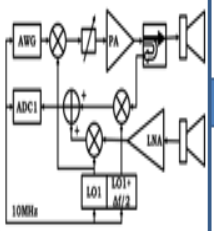
Software defined platform design

Unbiased study
Simulation/Experiment

Parallel Architecture

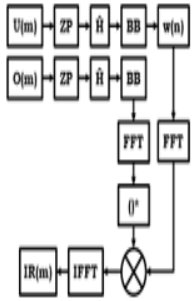


Frequency-interleaved Architecture

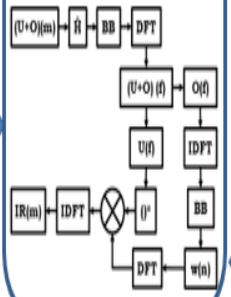


Hardware Architecture

Radix2-FFT algorithm



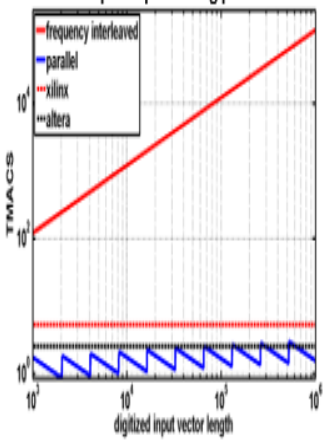
DFT Algorithm



Pulse compression Algorithm

Real-time processing is feasible with announced performances for the newest FPGA from Xilinx and Altera
➔ Architecture selected

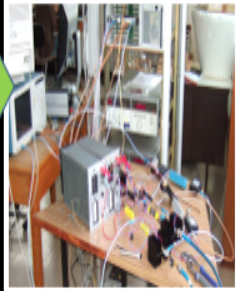
Required processing power



Intractable for real-time processing
➔ Architecture dismissed

Experimental setup

Software-defined radar prototype



Experimental ground



Comparison to state-of-the-art prototypes

platform	PANDORA [VanGenderen,98]	RCS meas. syst. [Paichard,07]	SAR imaging and telecom. [Garmatyuk,09]	IDROMel [IDROMel,09]	This work
Instantaneous bandwidth	776 MHz	800 MHz	500 MHz	20 MHz	800 MHz
Range resolution	0.19 m	N/A	0.3 m	N/A	0.1875 m
Tested range	N/A	10 m	1.5 to 5 m	N/A	60 m
Sampling scheme	Shannon	Sub-Nyquist + Bandpass	Shannon	Shannon	Bandpass
Sampling frequency	N/A	1.35 GS/s	1 GS/s	N/A	2 GS/s
Frequency Tuning Range	8 to 12 GHz	10 to 11.6 GHz	7 to 8 GHz	0.4 to 7.5 GHz	10 to 11.6GHz
Architecture	Super-heterodyne + stretch processing	Super-heterodyne + frequency-interleaving	Super-heterodyne	Super-heterodyne + 4x4 MIMO + I/Q channels	Super-heterodyne
Supported waveforms	Stepped Multitones, multiband FMCW	Multitones	Multitones	UMTS, GSM, IEEE.802.11/16	any
Pulse width	3.125 ms per step	100 to 200 ns	128 to 512 ns	Dependent on standard	500 ns up to 1 ms

The Experimental data is recorded on a hard drive and processed offline with MATLAB

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Results/Conclusion

Studied Waveforms

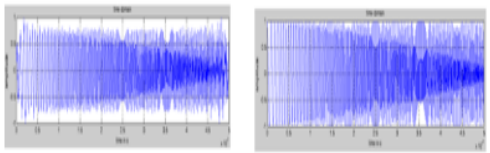
P3 phase-coded Multitones

$$MT(t)=\sum_{k=0}^{K-1}\exp\left(i\cdot 2\cdot\pi\cdot(k_0+k)\cdot\delta f\cdot t+\phi_k\right)$$

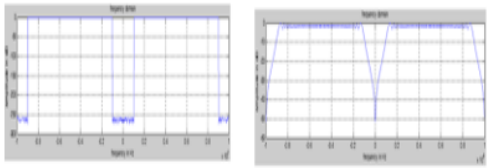
Linear Chirp

$$C(t)=\exp\left(i\cdot 2\cdot\pi\cdot\left(f_1+\frac{B}{2\cdot T}\cdot t\right)\cdot t\right)$$

Time Domain



Frequency Domain



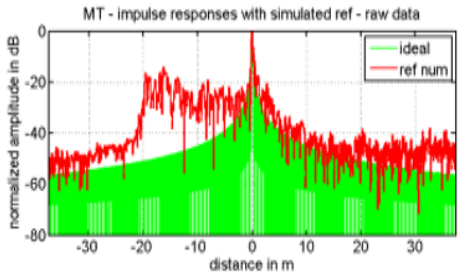
Tested configurations cover most radar applications from surveillance radar to high resolution radar

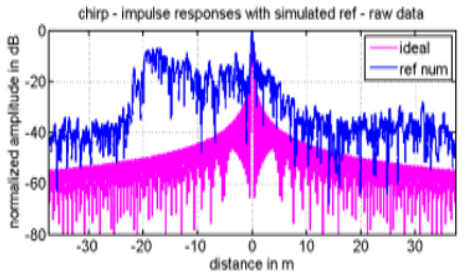
- Bandwith-time product > 75
- Tested Bandwidth → 1MHz, 10MHz, 150MHz, 800MHz
- Test Signal period → 500ns, 5µs, 50µs, 500µs, 1ms

Simulations vs Measurements

Parameters obtained in simulations, relative error for mainlobe width & sidelobe positions and difference in sidelobes amplitudes

Bandwidth	1 MHz	10 MHz	150 MHz	800 MHz
3 dB Mainlobe width	133m <1.9 %	13.3m <1.8 %	0.9m <2.3 %	0.225m <37 %
Sidelobe amplitudes	-13.27dB <0.3 dB	-13.27dB <0.3 dB	-13.27dB <0.3 dB	-13.27dB [-7;3 dB]
Sidelobe positions	±214.8 m <0.7 %	±21.4 m <1.7 %	±1.425 m < 3.1 %	±0.3 m <67 %





Detection range

Chirp yields an extended detection range compared to P3 phase-coded Multitones

simulations 13-17%
measurements 5-19%

Conclusions and perspectives

Software defined radar prototype

- Allows unbiased study of various waveforms
- Performances comparable to state-of-the-art prototypes

Chirp vs P3 phase coded Multitones

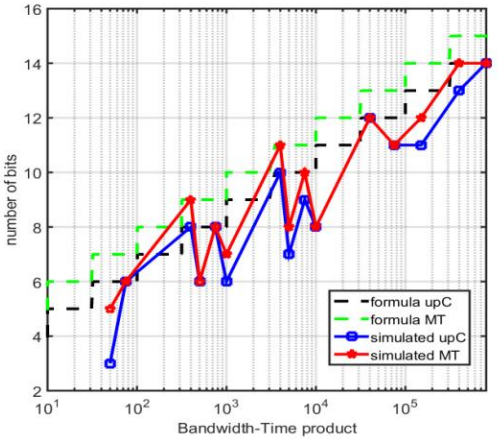
- Same performances in range profile
- 15% extra detection range with Chirp

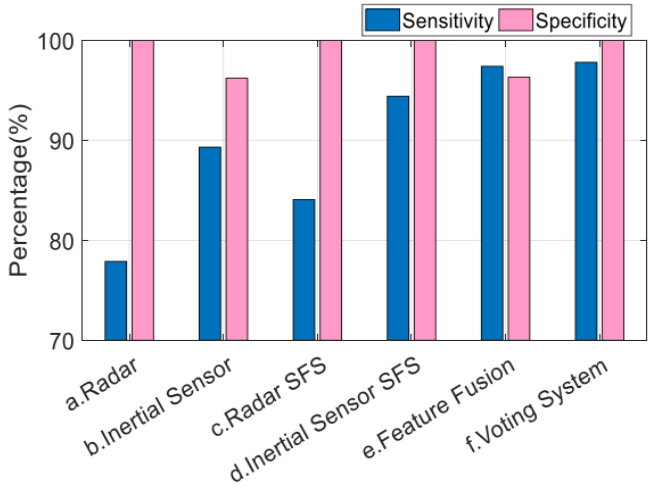
Future multifunction radar

- Multitones can incroporate telecommunications
- At least 15% loss in detection range for modulation with similar PMEPR reduction capability as P3 codes
- Multitones is the optimal waveform for overall performances (comm+radar)

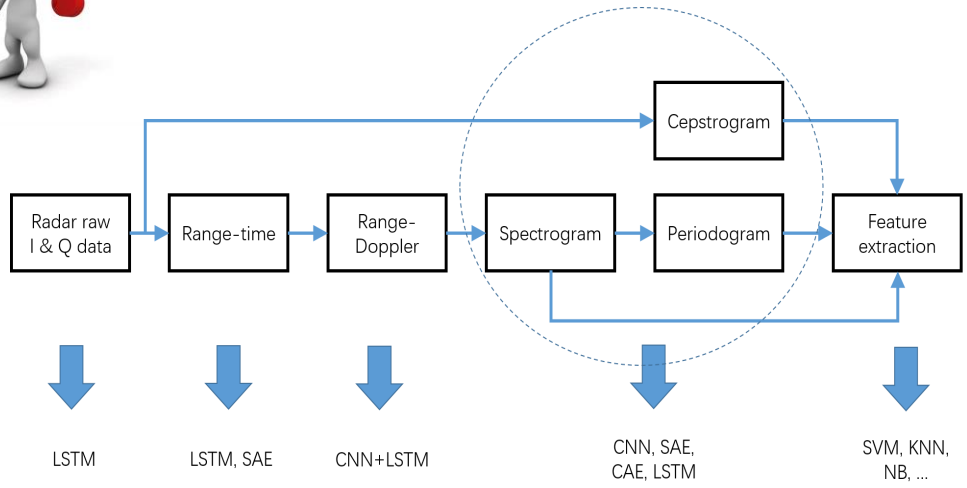
Presenting research data

	IMU		Radar	
Method	Accuracy(%)	Features #	Accuracy(%)	Features #
Fscore SVM	90.7	73	78.8	17
Fscore KNN	88.2	76	74.1	17
ReliefF SVM	91.1	164	74	20
ReliefF KNN	89.3	58	67	18
SFS SVM	95.6	35	85.6	20
SFS KNN	88.25	69	79.8	19

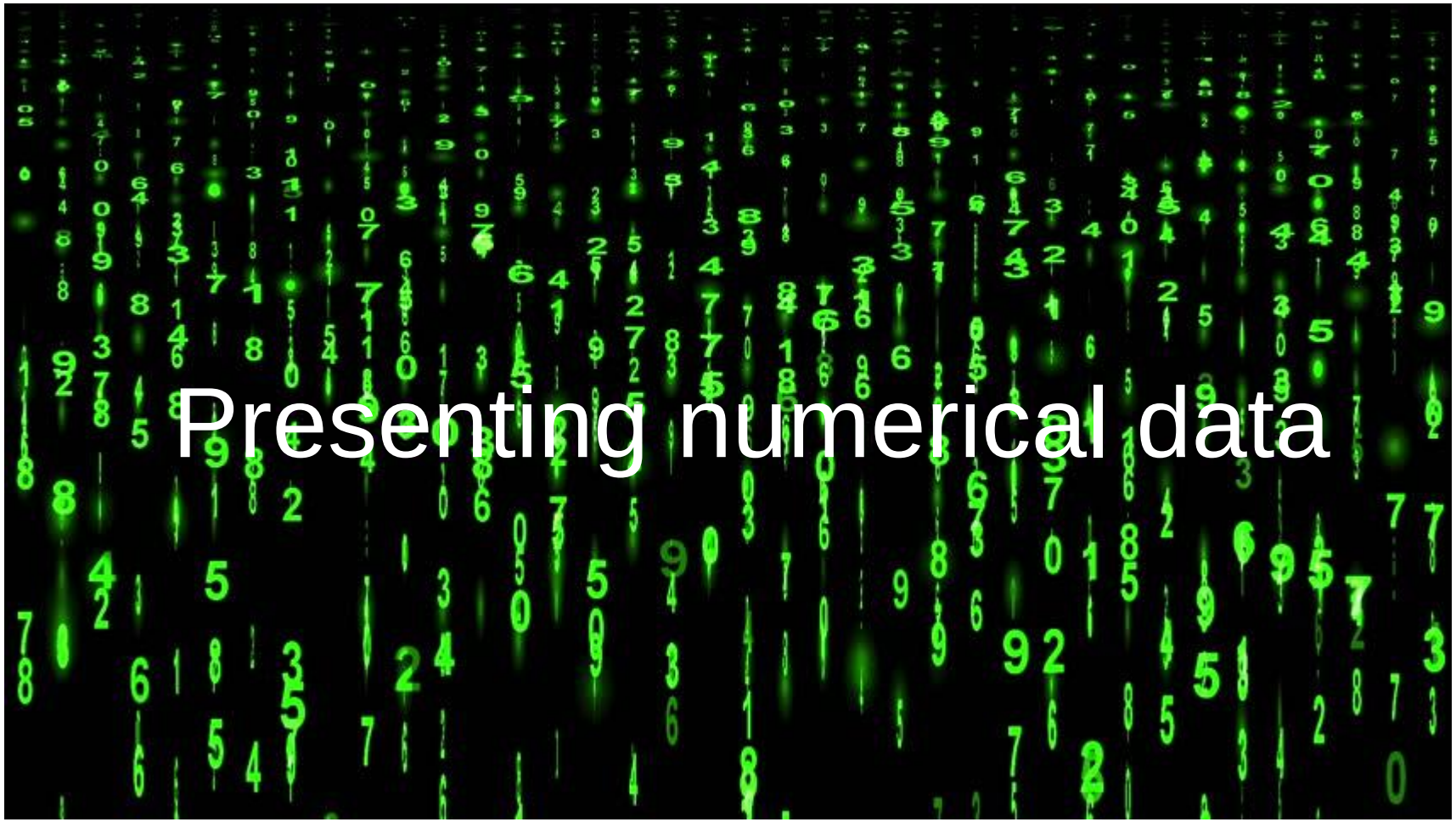








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Presenting numerical data

Presenting numerical data

- Essentially, you need to make the data tell a story or help support your story
  - It can be incorporated in main body of text as part of a sentence
  - It can be shown in tabulated form (table)
  - It can be presented in a graph

- The choice very much depends on the data:
  - How much?
  - How complex?

- Your job is to make it accessible to the reader, that's called translation – i.e. make something complicated understandable by most.

- Tables and graphs need to have descriptive captions and accompanying text to help interpret the data being shown

Don't bog down your audience with too much data.

Presenting numerical data

- You can incorporate numbers in text when you have only a few numbers to compare.
- Example:
- *The world health organization [6] reports that approximately 28-35% of people aged of 65 and over fall each year increasing to 32-42% for those over 70 years of age. The frequency of falls increases with age and frailty level. Frailty is accentuated by multi-morbidity and stroke recovery. Older people who are living in nursing homes fall more often than those who are living in community. Approximately 30-50% of people living in long-term care institutions fall each year, and 40% of them experienced recurrent falls.*

Presenting numerical data

- It is clearer than:
- Example:
- *The world health organization [6] reports that falls occur each year in:*
  - *65 years or over 28-35%*
  - *over 70 years 32-42%*
  - *Long term care 30-50%*
- *The frequency of falls increases with age and frailty level. Frailty is accentuated by multi-morbidity and stroke recovery. Older people who are living in nursing homes fall more often than those who are living in community. 40% of long-term care patients experienced recurrent falls.*

Numbers are generally given as digits rather than spelt out in the main body. Although that may differ between publications, so always check conventions.

Presenting numerical data

- Tables are very effective to present numerical data:
  - When you want to show how a single category of information varies when measured at different points (time or space)

Paris

	F°		C°		Rainfall	
	Average Daily		Average Daily		Average Monthly	
	MAX	MIN	MAX	MIN	IN	MM
Jan	43	34	6	1	2.2	56
Feb	45	34	7	1	1.8	46
March	54	39	12	4	1.4	35
April	60	43	16	6	1.7	42
May	68	49	20	10	2.2	57
June	73	55	23	13	2.1	54
July	76	58	25	15	2.3	59
Aug	75	58	24	14	2.5	64
Sept	70	53	21	12	2.2	55
Oct	60	46	16	8	2.0	50
Nov	50	40	10	5	2.0	51
Dec	44	36	7	2	2.0	50

Presenting numerical data

- Tables are very effective to present numerical data:
  - When the data contains relatively few numbers.
  - This is very hard for the audience to assimilate in a short time a big slab of numbers especially in presentations when they have little time to look at it or review it.

Paris

	F°		C°		Rainfall	
	Average Daily		Average Daily		Average Monthly	
	MAX	MIN	MAX	MIN	IN	MM
Jan	43	34	6	1	2.2	56
Feb	45	34	7	1	1.8	46
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Aug	75	58	24	14	2.5	64
Sept	70	53	21	12	2.2	55
Oct	60	46	16	8	2.0	50
Nov	50	40	10	5	2.0	51
Dec	44	36	7	2	2.0	50

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual Average
1958	-99.99	-99.99	315.7	317.45	317.5	317.26	315.86	314.93	313.2	312.44	313.33	314.67	-99.99
1959	315.62	316.38	316.71	317.72	318.29	318.16	316.54	314.8	313.84	313.26	314.8	315.58	315.98
1960	316.43	316.97	317.58	319.02	320.03	319.59	318.18	315.91	314.16	313.84	315	316.19	316.91
1961	316.93	317.7	318.54	319.48	320.58	319.77	318.57	316.79	314.8	315.38	316.1	317.01	317.64
1962	317.94	318.56	319.68	320.63	321.01	320.55	319.58	317.4	316.25	315.42	316.69	317.7	318.45
1963	318.74	319.08	319.86	321.39	322.24	321.47	319.74	317.77	316.21	315.99	317.12	318.31	318.99
1964	319.57	-99.99	-99.99	-99.99	322.24	321.89	320.44	318.7	316.7	316.79	317.79	318.71	-99.99
1965	319.44	320.44	320.89	322.13	322.16	321.87	321.39	318.8	317.81	317.3	318.87	319.42	320.04
1966	320.62	321.59	322.39	323.87	324.01	323.75	322.39	320.37	318.64	318.1	319.79	321.08	321.38
1967	322.06	322.5	323.04	324.42	325	324.09	322.55	320.92	319.31	319.31	320.72	321.96	322.16
1968	322.57	323.15	323.89	325.02	325.57	325.36	324.14	322.03	320.41	320.25	321.31	322.84	323.05
1969	324	324.42	325.64	326.66	327.34	326.76	325.88	323.67	322.38	321.78	322.85	324.12	324.63
1970	325.03	325.99	326.87	328.14	328.07	327.66	326.35	324.69	323.1	323.16	323.98	325.13	325.68
1971	326.17	326.68	327.18	327.78	328.92	328.57	327.34	325.46	323.36	323.57	324.8	326.01	326.32
1972	326.77	327.63	327.75	329.72	330.07	329.09	328.05	326.32	324.93	325.06	326.5	327.55	327.45
1973	328.55	329.56	330.3	331.5	332.48	332.07	330.87	329.31	327.51	327.18	328.16	328.64	329.68
1974	329.35	330.71	331.48	332.65	333.09	332.25	331.18	329.4	327.43	327.37	328.46	329.57	330.25
1975	330.4	331.41	332.04	333.31	333.96	333.6	331.91	330.06	328.56	328.34	329.49	330.76	331.15
1976	331.75	332.56	333.5	334.58	334.87	334.34	333.05	330.94	329.3	328.94	330.31	331.68	332.15
1977	332.93	333.42	334.7	336.07	336.74	336.27	334.93	332.75	331.59	331.16	332.4	333.85	333.9
1978	334.97	335.39	336.64	337.76	338.01	337.89	336.54	334.68	332.76	332.55	333.92	334.95	335.51
1979	336.23	336.76	337.96	338.89	339.47	339.29	337.73	336.09	333.91	333.86	335.29	336.73	336.85
1980	338.01	338.36	340.08	340.77	341.46	341.17	339.56	337.6	335.88	336.02	337.1	338.21	338.69
1981	339.23	340.47	341.38	342.51	342.91	342.25	340.49	338.43	336.69	336.86	338.36	339.61	339.93
1982	340.75	341.61	342.7	343.57	344.13	343.35	342.06	339.81	337.98	337.86	339.26	340.49	341.13
1983	341.37	342.52	343.1	344.94	345.75	345.32	343.99	342.39	339.86	339.99	341.15	342.99	342.78
1984	343.7	344.5	345.28	347.08	347.43	346.79	345.4	343.28	341.07	341.35	342.98	344.22	344.42
1985	344.97	346	347.43	348.35	348.93	348.25	346.56	344.68	343.09	342.8	344.24	345.55	345.9
1986	346.3	346.96	347.86	349.55	350.21	349.54	347.94	345.9	344.85	344.17	345.66	346.9	347.15
1987	348.02	348.47	349.42	350.99	351.84	351.25	349.52	348.1	346.45	346.36	347.81	348.96	348.93
1988	350.43	351.73	352.22	353.59	354.22	353.79	352.38	350.43	348.72	348.88	350.07	351.34	351.48
1989	352.76	353.07	353.68	355.42	355.67	355.13	353.9	351.67	349.8	349.99	351.29	352.52	352.91
1990	353.66	354.7	355.39	356.2	357.16	356.23	354.82	352.91	350.96	351.18	352.83	354.21	354.19
1991	354.72	355.75	357.16	358.6	359.33	358.24	356.17	354.02	352.15	352.21	353.75	354.99	355.59
1992	355.98	356.72	357.81	359.15	359.66	359.25	357.02	355	353.01	353.31	354.16	355.4	356.37
1993	356.7	357.16	358.38	359.46	360.28	359.6	357.57	355.52	353.69	353.99	355.34	356.8	357.04
1994	358.37	358.91	359.97	361.26	361.68	360.95	359.55	357.48	355.84	355.99	357.58	359.04	358.89
1995	359.97	361	361.64	363.45	363.79	363.26	361.9	359.46	358.05	357.76	359.56	360.7	360.88
1996	362.05	363.25	364.02	364.72	365.41	364.97	363.65	361.48	359.45	359.6	360.76	362.33	362.64
1997	363.18	364	364.56	366.35	366.79	365.62	364.47	362.51	360.19	360.77	362.43	364.28	363.76
1998	365.33	366.15	367.31	368.61	369.3	368.87	367.64	365.77	363.9	364.23	365.46	366.97	366.63
1999	368.15	368.87	369.59	371.14	371	370.35	369.27	366.93	364.63	365.13	366.67	368.01	368.31
2000	369.14	369.46	370.52	371.66	371.82	371.7	370.12	368.12	366.62	366.73	368.29	369.53	369.48
2001	370.28	371.5	372.12	372.87	374.02	373.3	371.62	369.55	367.96	368.09	369.68	371.24	371.02
2002	372.43	373.09	373.52	374.86	375.55	375.41	374.02	371.49	370.7	370.25	372.08	373.78	373.1
2003	374.68	375.63	376.11	377.65	378.35	378.13	376.62	374.5	372.99	373.01	374.35	375.7	375.64
2004	376.79	377.37	378.41	380.52	380.63	379.57	377.79	375.86	374.07	374.24	375.86	377.47	377.38
2005	378.37	379.69	380.41	382.1	382.28	382.13	380.66	378.71	376.42	376.88	378.32	380.04	379.67
2006	381.38	382.03	382.64	384.62	384.95	384.06	382.29	380.47	378.67	379.06	380.14	381.74	381.84
2007	382.45	383.68	384.23	386.26	386.39	385.87	384.39	381.78	380.73	380.81	382.33	383.69	383.55
2008	385.07	385.72	385.85	386.71	388.45	387.64	386.1	383.95	382.91	382.73	383.96	385.02	385.34

Values above represent monthly concentrations adjusted to represent 2400 hours on the 15th day of each month. Units are parts per million by volume (ppmv) expressed in the 2003A SIO manometric mole fraction scale. The "annual average" is the arithmetic mean of the twelve monthly values where no monthly values are missing.

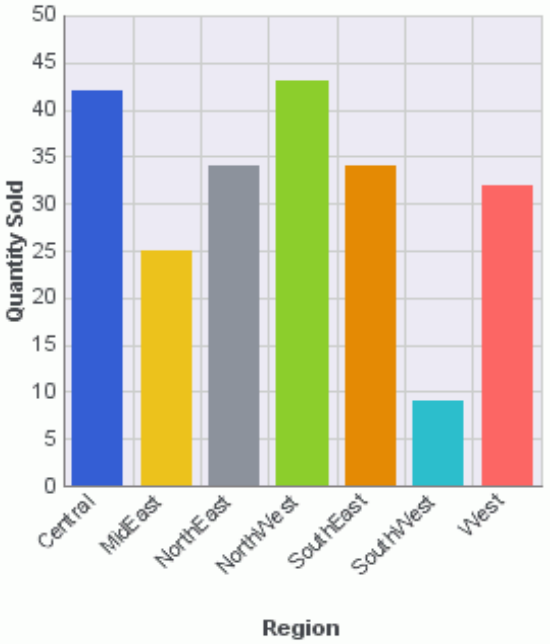
Presenting numerical data

- Tables are very effective to present numerical data:
  - When the precise value is crucial to your argument and a graph would not convey the same level of precision.

Use a Table to:
Look up individual values.
Compare individual values.
Use values precisley.

Degree Type	Customer Income Range
Associate Degree	A: $0 TO $24,999
	B: $25,000 TO $49,999
	C: $50,000 TO $74,999
	D: $75,000 TO $99,999
	E: $100,000 OR MORE
Bachelor's Degree	B: $25,000 TO $49,999
	C: $50,000 TO $74,999
	D: $75,000 TO $99,999
	E: $100,000 OR MORE
Master's Degrees	A: $0 TO $24,999
	B: $25,000 TO $49,999
	C: $50,000 TO $74,999
	D: $75,000 TO $99,999
	E: $100,000 OR MORE

Use a Chart to:
Show overall shape of values.
Show relationship among multiple values using patterns and trends.

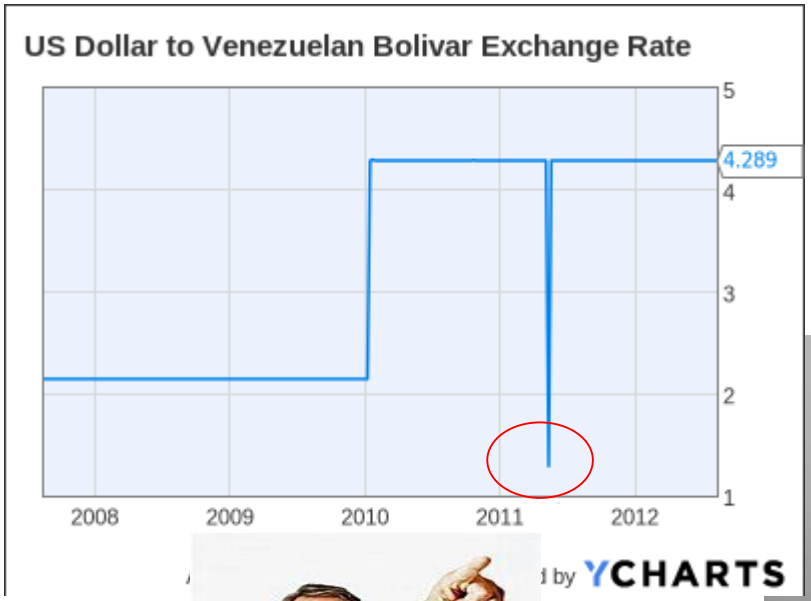




Presenting numerical data

- Tables are very effective to present numerical data:
  - When you don't want outliers (very high or low numbers) to detract from the message contained in the rest of the dataset.

Year	Exchange rate US-> Bolivar
2008	2.211
2009	2.211
2010	4.289
2011	4.289
2012	4.289







Presenting numerical data

- Table design: How to make a table clear and understandable
  - Columns and rows
    - **Most people find it easier to identify a pattern in a column.**
    - **Arrange your table so that the pattern you want to highlight appears in a column**
    - **It is also easier to interpret data if they are arranged in order of magnitude, although not always feasible.**
  - Several columns or categories of information
    - **This may render a table difficult to interpret**
    - **So identify which column contains the most important trend and this should be used to arrange your table**
    - **If columns are equally important, it might be better to have more simple tables rather than a big complex one.**

Presenting numerical data

- Table design: How to make a table clear and understandable
  - Numbers in tables
    - **Should be presented in a simple format**
      - Consider that 2 numbers after the decimal point are More than enough. So don't be afraid to round numbers.

        pi = 3.14159265358979323846264338327950288419716939937510

        pi ≈ 3.14

        $5,700,000 or $5,7 million

        0.03615 or $3.61 \cdot 10^{-2}$

        what is easier?
  - All tables should have a caption that is sufficient descriptive so that the reader can read the table and understand the content without reading the text. If the data comes from a book, journal or … **reference it, cite it.** Give information about your data collection if you carried it yourself

Presenting numerical data

- Table design: How to make a table clear and understandable
  - When more than one table is presented, it is usual to give each one a unique number like Table I or Table II so they can be easily referred to in the main body. Furthermore, you'll be able to create a table of tables for a large document using word processors.

Table of tables

Table 1: Advantages and drawbacks of multitones and OFDM 2-10
Table 2: New radar concepts and implications in digital and analogue parts 2-16
Table 3: IDROMel basic features 3-5
Table 4: Synthesis of the known OFDM platform features 3-7
Table 5: linearization techniques and their improvement capabilities and drawbacks 3-10
Table 6: linearization techniques and their compensation capabilities and system implications (49) 3-11
Table 7: evaluation of linearization techniques based on components, input power, waveform and performance criteria improvements 3-13
Table 8: simulated improvements for ADC with multicarrier signals 3-14

Presenting numerical data

- Table design: How to make a table clear and understandable
  - The formatting of the table should not resemble a spreadsheet where each entry is bounded by a box since this makes it difficult to read across rows or down columns.
  - However, the design of the table should help the reader interpret the data and so the use of lines and/or bold text to separate headings from the body of data, or highlighting/shading specific rows or may be effective.
  - Avoid large gaps between columns since this also makes it difficult to read along a row.

Presenting numerical data

- **Examples of poor and better practice in the presentation of data in tables**
  - Poor example

Region	% adults taking a holiday
Ain	50
Bourgogne	64
Bretagne	56
Dordogne	42
Ile de France	30
Normandie	41
Hauts de France	64

Can you identify what is wrong here?

Presenting numerical data

- No title
- Where does the data come from?
- Row titles straddle two lines
- Each bounded as if in a spreadsheet
- The alphabetical listing is not order showing an order in magnitude

Region	% adults taking a holiday
Ain	50
Bourgogne	64
Bretagne	56
Dordogne	42
Ile de France	30
Normandie	41
Hauts de France	64

Presenting numerical data

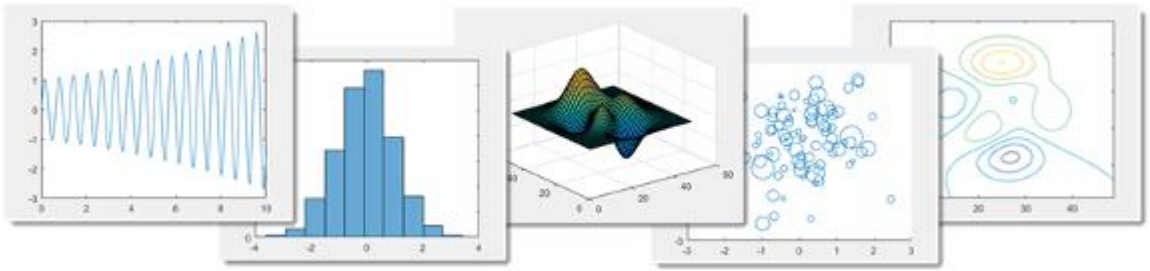
- Better example

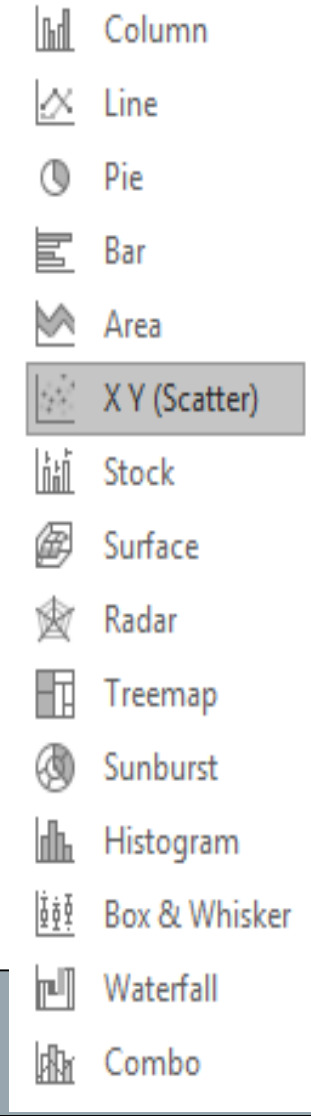
Table 1: Regional Differences in the Percentage of Adults Taking a Holiday in 2017 [ref]

Region	%
Ile de France	30
Normandie	41
Dordogne	42
Ain	50
Bretagne	56
Hauts de France	64
Bourgogne	64

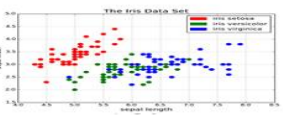
Presenting numerical data

- Graphs
  - Good to describe, explore or summarise numerical data because the use of an image can simplify complex information and help highlight patterns and trend in data.
  - More effective for large amount of data but can also be used for small amount of data
  - There are many types of graphs to choose from:

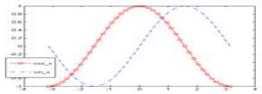


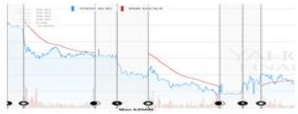


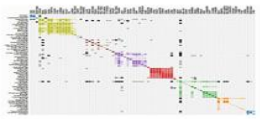
TYPES OF GRAPHS

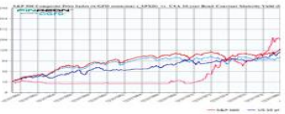

Iris

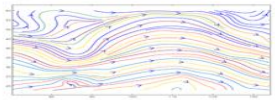

Candlestick

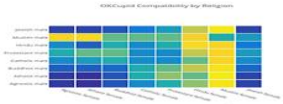

Legend

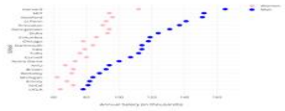

Box Annotation


Les Mis


Stocks

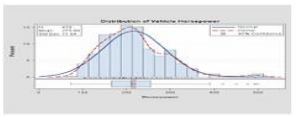

Streamline

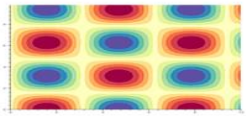

Categorical Heat Map

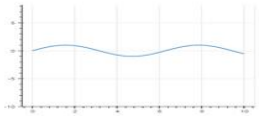

Categorical Dot Plot

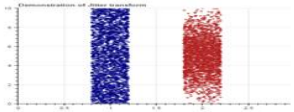

Unemployment

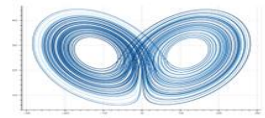

Periodic

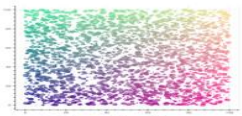

Histogram


Image

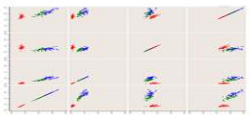

Slider

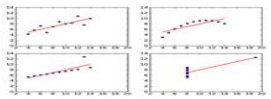

Jitter

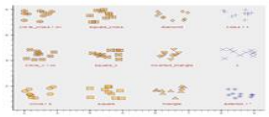

Lorenz

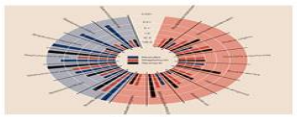

Color Scatter

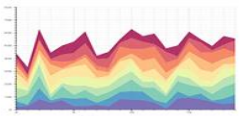

Color Sliders

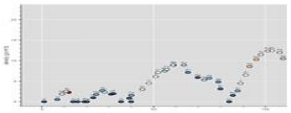

Iris Splom

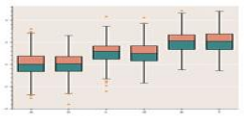

Anscombe

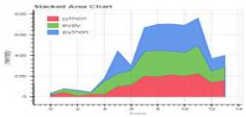

Markers

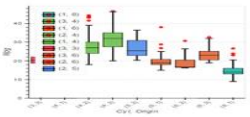

Burtin

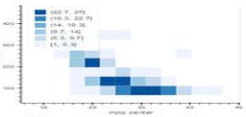

Brewer


Elements

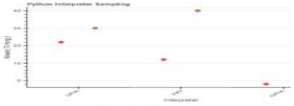

Boxplot

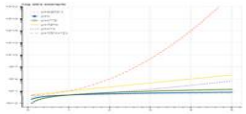

Area Chart

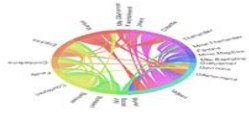

Boxplot Chart

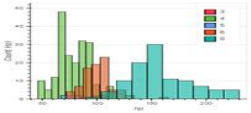

Heatmap Chart

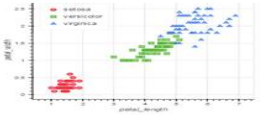

Donut Chart

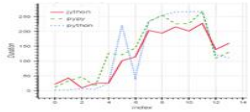

Dot Chart

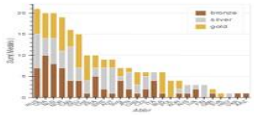

Logaxis

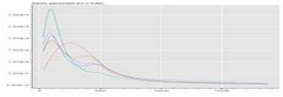

Chord Chart

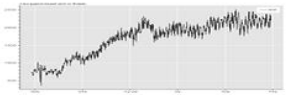

Histograms Chart

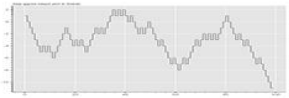

Iris Scatter

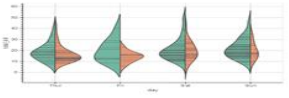

Line Chart

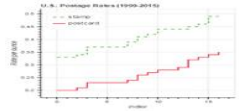

Stacked Bar Chart

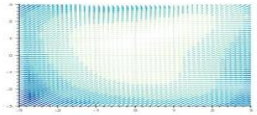

GGplot Density

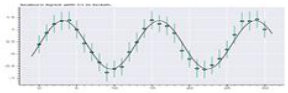

GGplot Line

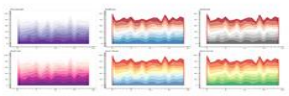

GGplot Step

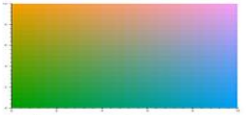

Seaborn Violin


Step Chart


Quiver


Seaborn Sinerror


Palettes


Image RGBA

Presenting numerical data

Subfigure labels or subtitle

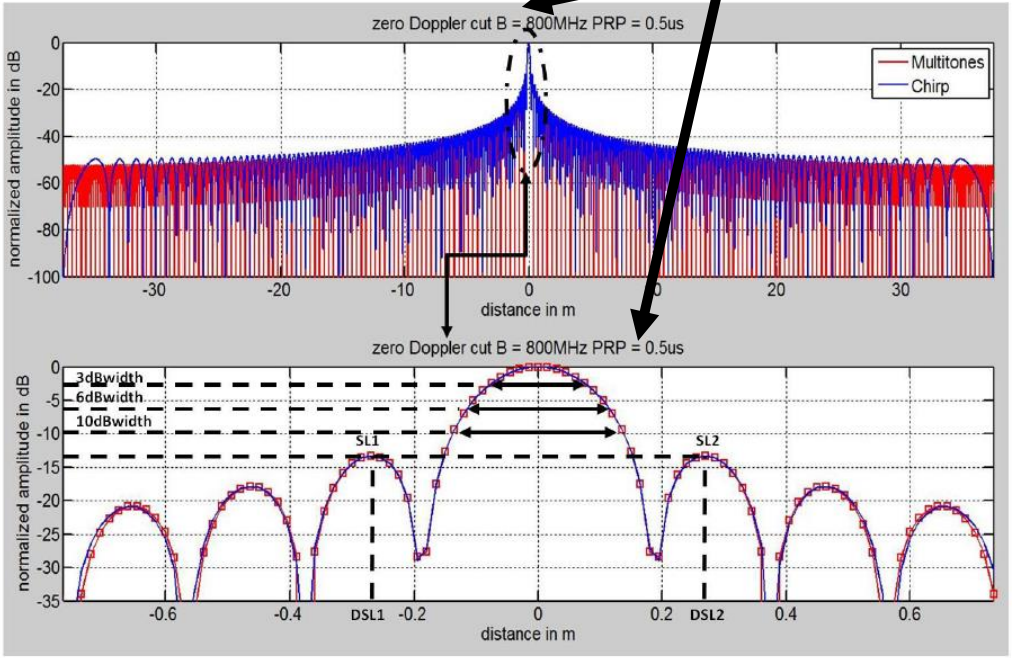


Figure 38: pulse compression measurements in distance for multitones and chirp when PRP = 500ns and B = 800MHz

Detailed caption with identifiable number

Presenting numerical data

Axis labels and UNITS

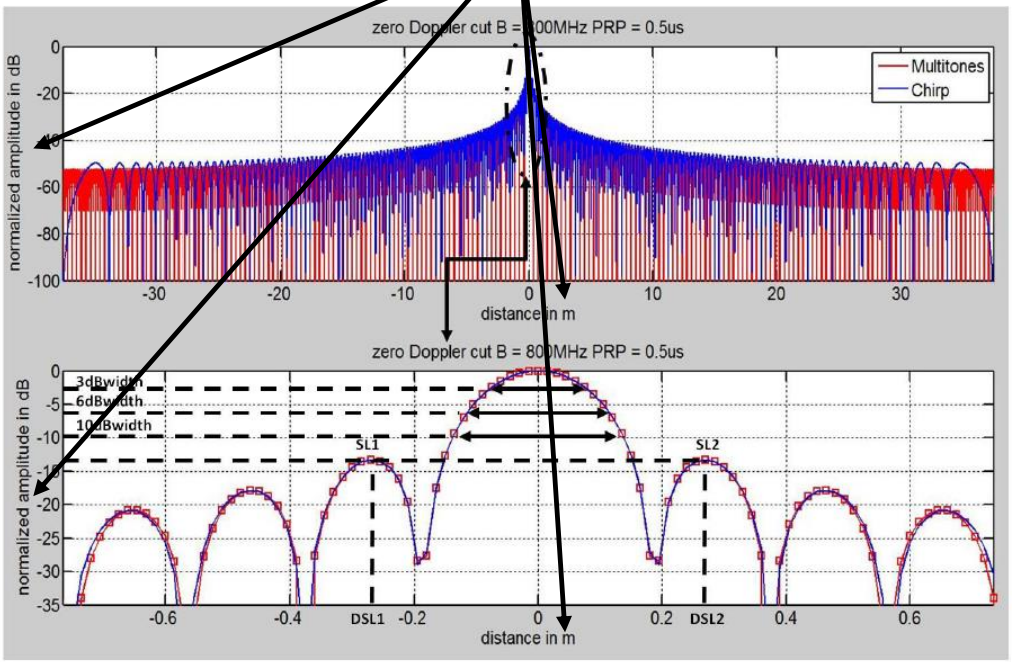


Figure 38: pulse compression measurements in distance for multitones and chirp when PRP = 500ns and B = 800MHz

Presenting numerical data

A legend

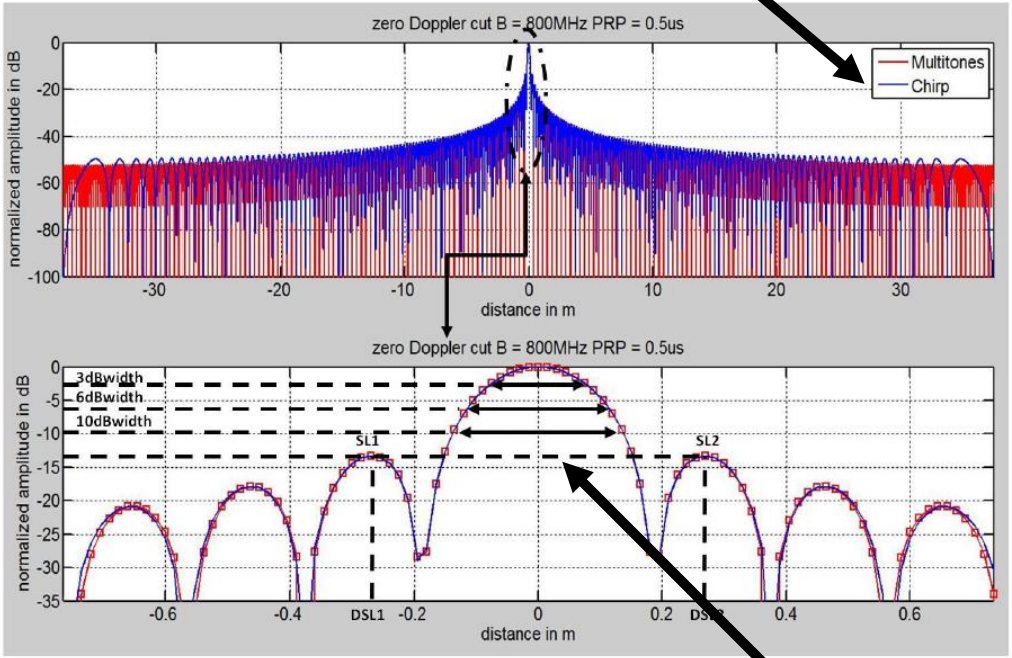


Figure 38: pulse compression measurements in distance for multitones and chirp when PRP = 500ns and B = 800MHz

Gridlines if needed if it helps read values

Highlights if needed to help interpret data

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Presenting research data



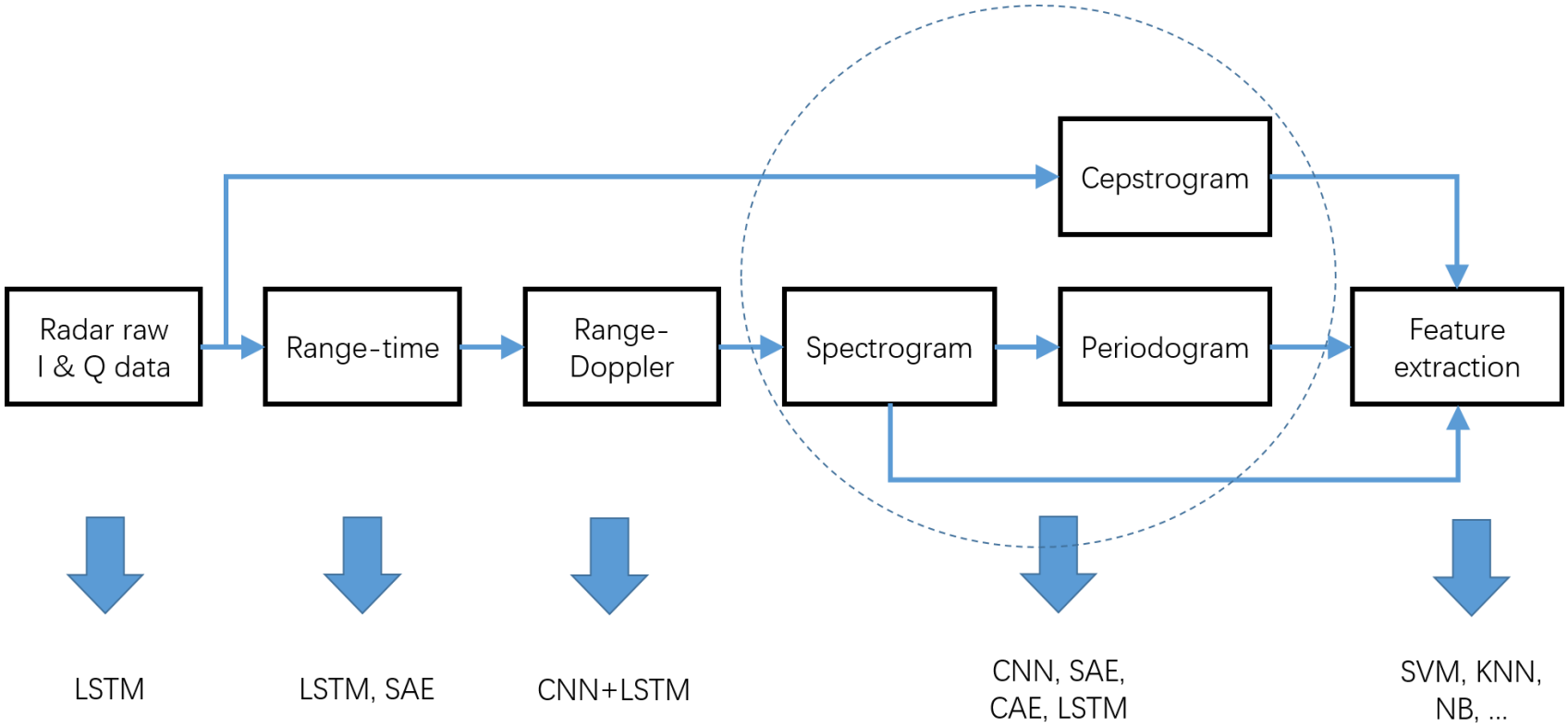
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- [49] S. Song *et al.*, “Multimodal Multi-Stream Deep Learning for Egocentric Activity Recognition,” in *2016 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2016, pp. 378–385.
- [50] R. Saeedi, S. Norgaard, and A. H. Gebremedhin, “A closed-loop deep learning architecture for robust activity recognition using wearable sensors,” in *2017 IEEE International Conference on Big Data (Big Data)*, 2017, pp. 473–479.
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- [53] Y. Kim and B. Toomajian, “Hand Gesture Recognition Using Micro-Doppler Signatures With Convolutional Neural Network,” *IEEE Access*, vol. 4, pp. 7125–7130, 2016.
- [54] R. P. Trommel, R. I. A. Harmanny, L. Cifola, and J. N. Driessen, “Multi-target human gait classification using deep convolutional neural networks on micro-doppler spectrograms,” in *2016 European Radar Conference (EuRAD)*, 2016, pp. 81–84.
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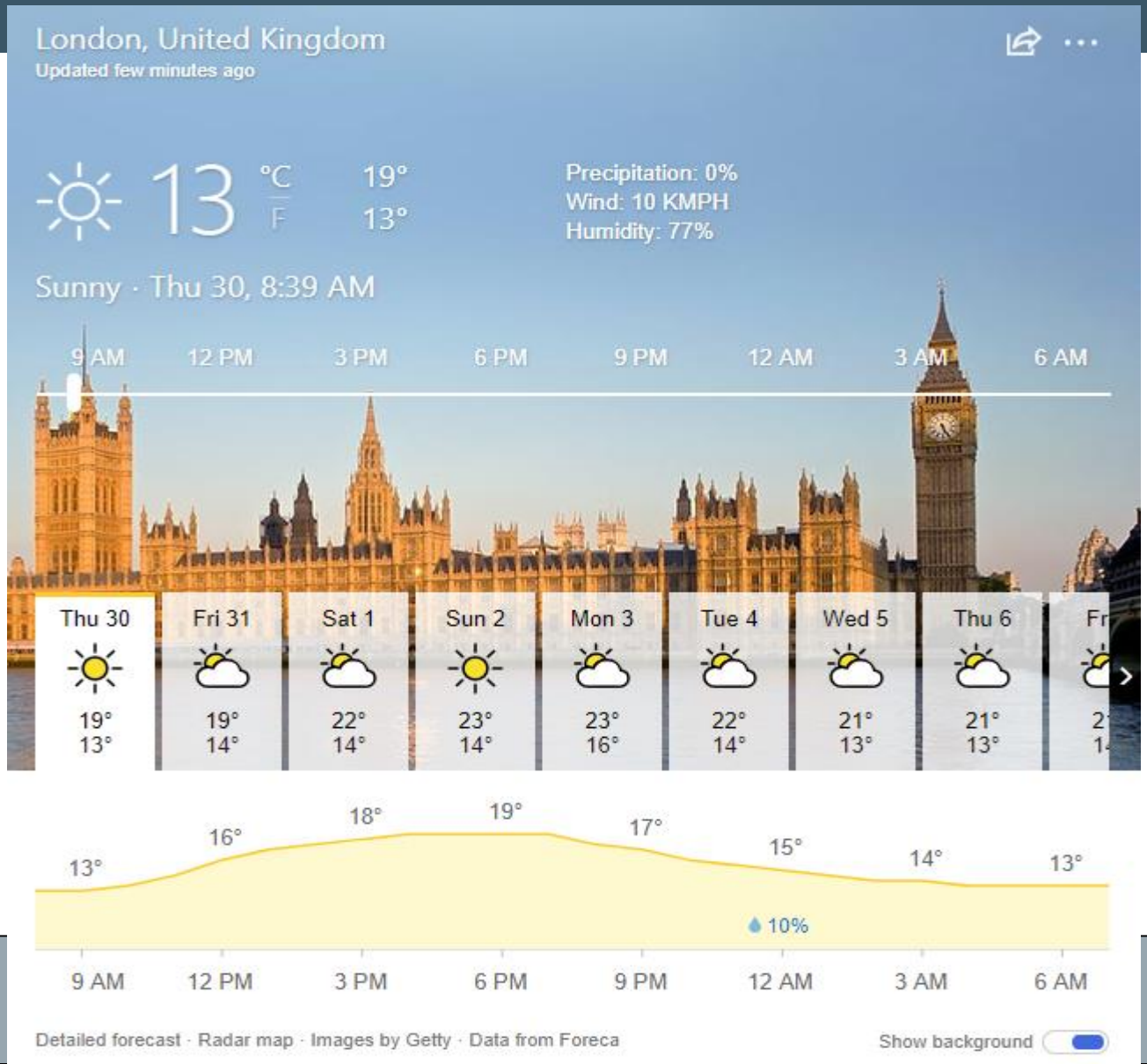
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Presenting research data



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Presenting research data



Presenting research data

Table I: comparison of the the prototype to the RF platforms from the literature review

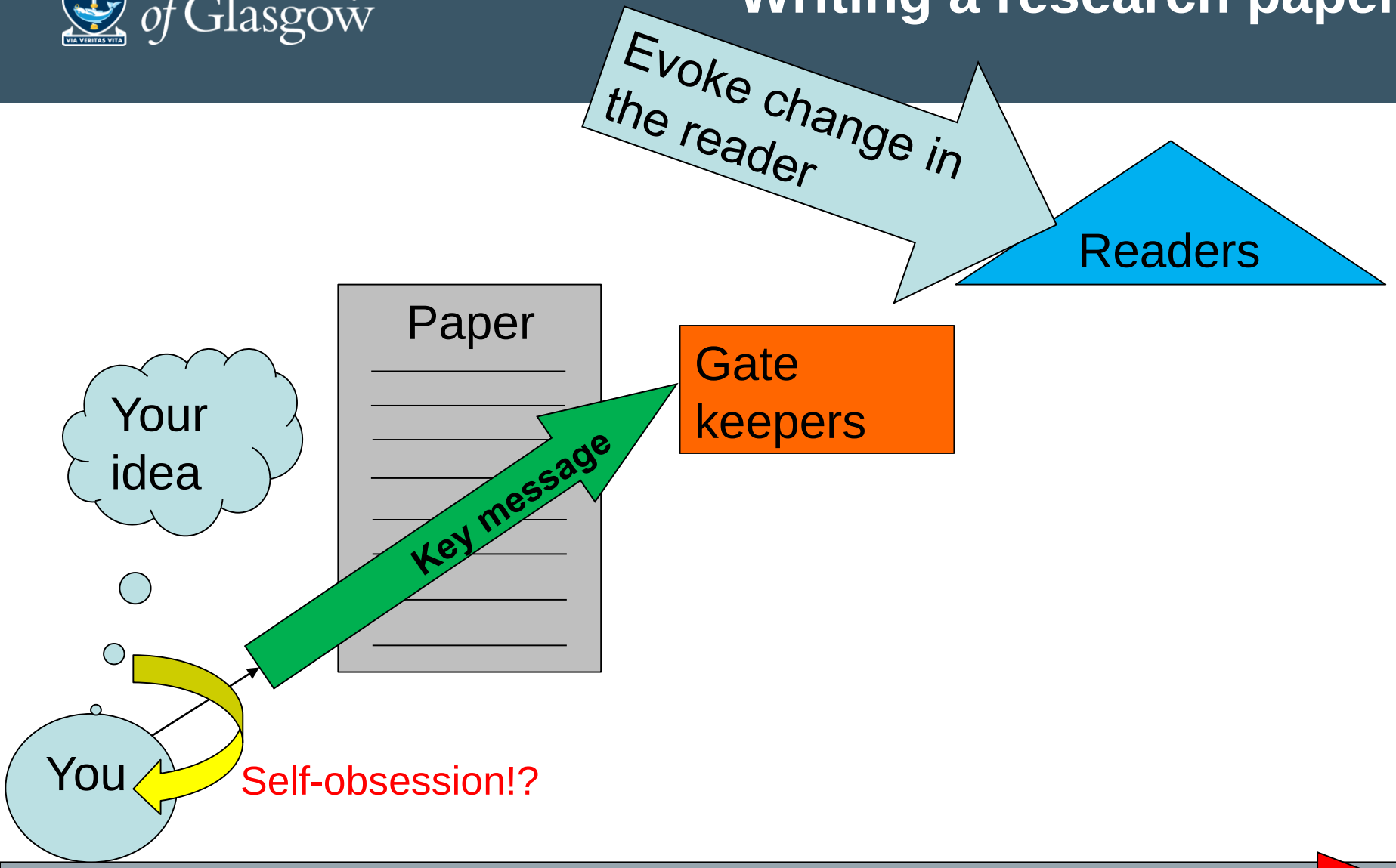
platform	PANDORA [10]	RCS meas. syst. [11]	SAR imaging and telecom. [6]	IDROMel [12]	This work
Instantaneous bandwidth	776 MHz	800 MHz	500 MHz	20 MHz	800 MHz
Range resolution	0.19 m	N/A	0.3 m	N/A	0.1875 m
Tested range	N/A	10 m	1.5 to 5 m	N/A	60 m
Sampling scheme	Shannon	Sub-Nyquist + Bandpass	Shannon	Shannon	Bandpass
Sampling frequency	N/A	1.35 GS/s	1 GS/s	N/A	2 GS/s
Frequency Tuning Range	8 to 12 GHz	10 to 11.6 GHz	7 to 8 GHz	0.4 to 7.5 GHz	10 to 11.6GHz
Architecture	Super-heterodyne + stretch processing	Super-heterodyne + frequency-interleaving	Super-heterodyne	Super-heterodyne + 4x4 MIMO + I/Q channels	Super-heterodyne
Supported waveforms	Stepped Multitones, multiband FMCW	Multitones	Multitones	UMTS, GSM, IEEE.802.11/16	any
Pulse width	3.125 ms per step	100 to 200 ns	128 to 512 ns	Dependent on standard	500 ns up to 1 ms

Representing research data

- Essentially
  - Arrange the data in a way that the audience can understand complex data simply, don't make it simpler.
  - First, decide on the message to be conveyed and the purpose
  - The question is putting a hierarchy in the information going from the macro to micro level
    - **Show the data**
    - **Give an overview**
    - **Carry a message**
    - **Compare different aspects**
    - **….**

One thing is don't twist the data or make it say something it does not

Writing a research paper



Writing a research paper

- Know you have a message (a good story), then find the appropriate medium
- Simply cataloguing data is not enough



Writing a research paper

- What does the reader want?
  - Clarity
  - Consistency
  - Impact
  - Ease



Writing a research paper

- Is the paper consistent with …
  - The journal aims
  - The assignment
- Are the key points obvious?
- Do they support the purpose of the work?
- Is the work important?

www.unc.edu/~haipeng/teaching/sci.pdf

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Writing a research paper

- Is it readable!?
- “The smallest of the URF’s (URFA6L), a 207-nucleotide (nt) reading frame overlapping out of phase the NH_2-terminal portion of the adenosinetriphosphatase (ATPase) subunit 6 gene has been identified as the animal equivalent of the recently discovered yeast H^+-ATPase subunit 8 gene.”

www.unc.edu/~haipeng/teaching/sci.pdf

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Writing a research paper

- Is it readable!?
- “The smallest of the URF’s (URFA6L), a 207-nucleotide (nt) reading frame overlapping out of phase the NH_2-terminal portion of the adenosinetriphosphatase (ATPase) subunit 6 gene has been identified as the animal equivalent of the recently discovered yeast H^+-ATPase subunit 8 gene.”

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Writing a research paper

- Is it readable!?
- “The smallest of the URF’s (URFA6L), a 207-nucleotide (nt) reading frame overlapping out of phase the NH_2-terminal portion of the adenosinetriphosphatase (ATPase) subunit 6 gene; it has been identified as the animal equivalent of the recently discovered yeast H^+-ATPase subunit 8 gene.”

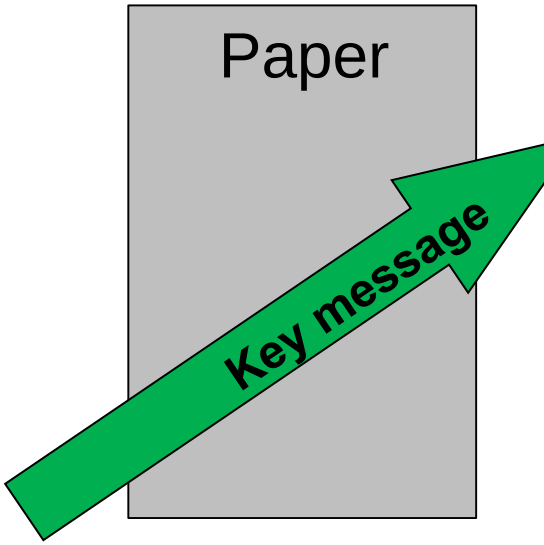
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Writing a research paper

- Is it readable!?
- "The smallest of the URF's (URFA6L), a 207-nucleotide (nt) reading frame overlapping out of phase the NH_2-terminal portion of the adenosinetriphosphatase (ATPase) subunit 6 gene; it has been identified as the animal equivalent of the recently discovered yeast H^+-ATPase subunit 8 gene."

Writing a research paper

- Is it readable!?
- “The smallest of the URF’s (URFA6L) has been identified as the animal equivalent of the recently discovered yeast H^+-ATPase subunit 8 gene.”



www.unc.edu/~haipeng/teaching/sci.pdf

Writing a research paper

- What does the gatekeeper care about
  - Good writing clarity and style
  - Useful title to paper
  - Guidelines to be followed

Noble: Studies in Higher Education 13 (1) 1989

Publish or Perish - what 23 Journal Editors have to say

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Writing a research paper

- Common mistakes
  - Guidelines not followed
  - Unprofessional appearance
  - Bad writing
  - Subject of no interest or out of scope
  - Poor stats, tables, figures
  - Not thorough
  - Too simple – reporting
  - Written at wrong level (not addressed to the right audience)

Noble: Studies in Higher Education 13 (1) 1989

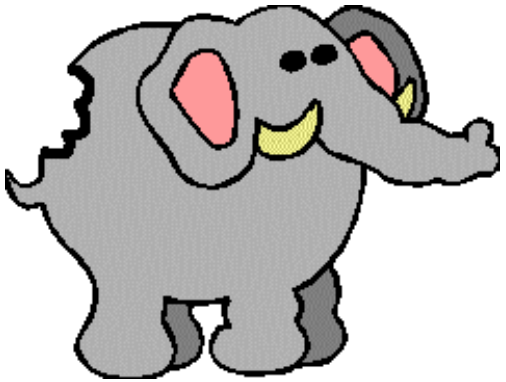
Publish or Perish - what 23 Journal Editors have to say

Writing a research paper

- Advice
  - Have the paper read by someone else for critique before submission
  - Think about what the audience wants to see not what you want to say!
  - Be a stickler for details(e.g. missing/inaccurate references are a sign of laziness)

Writing a research paper

- Getting started
  - How do you eat an elephant?
    - **Answer: one bite at a time**
  - Plan your paper
    - **Use a skeleton structure**
    - **A mindmap**



Writing a research paper

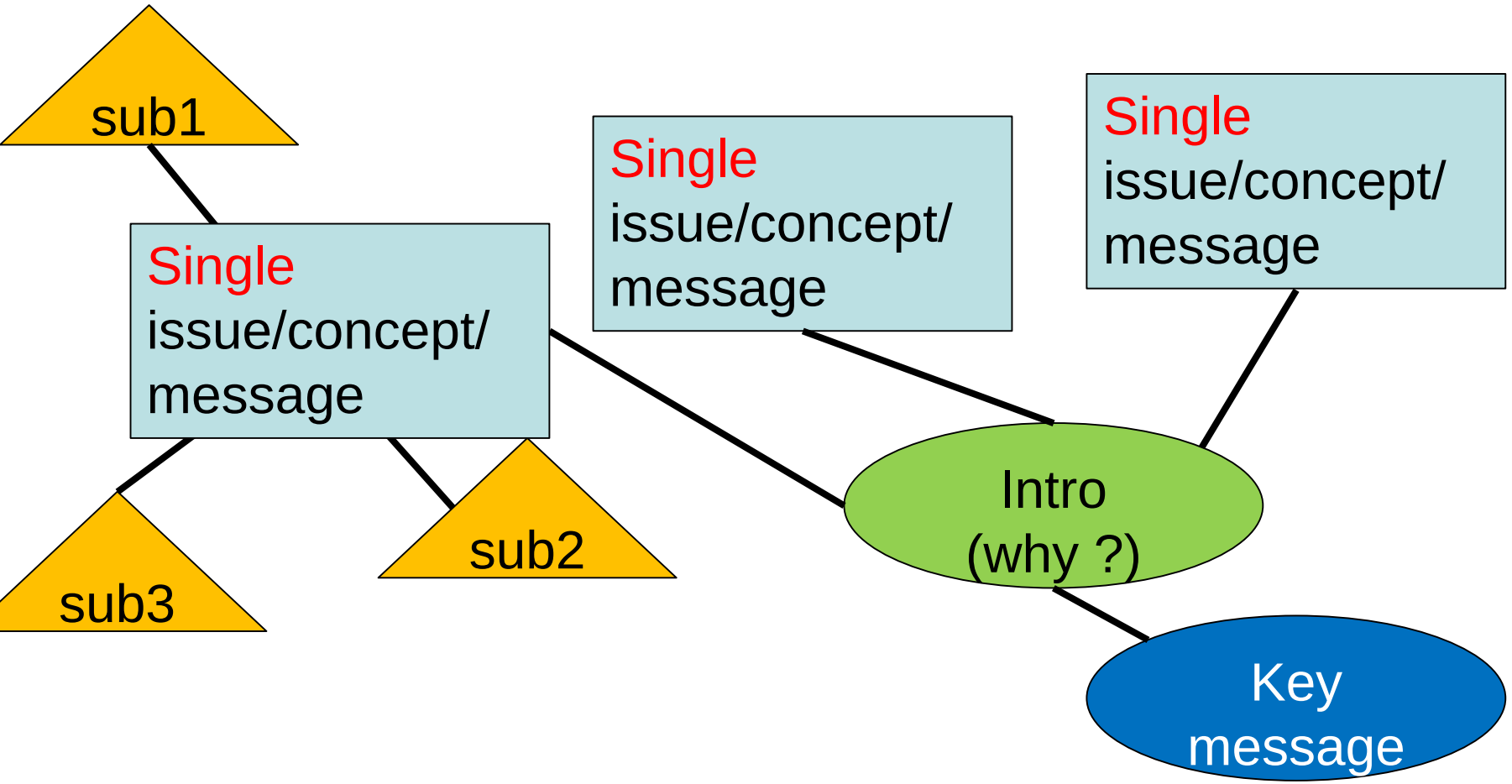
- Questions to get going…
  - Who am I writing this for ?
  - What am I doing ?
  - Why am I doing it ?
  - What has been done in the past ?
  - Who did it ?
  - What were the effects ?
  - Why was this unsatisfactory ?
  - What have I tried that worked ?
  - What else did I find ?

Most important question

- **What is the key benefit for readers? !**
  - Many writers consider this too late…

- Your answer provides a destination for the paper in every sense and helps structure…

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Think. Be ordered. Be ruthless. This could take some time.

Writing a research paper

- A reviewer's advice (marker)
  1. I read the title (so have a title)
  2. I read the abstract (is it interesting and well structured?)
  3. I skim through the intro

  90% of my opinion has been formed about your performance

  Then I read the rest of the paper to support my view.

Writing a research paper

- Title
  - Keep it simple and memorable
  - Use the fewest number of words possible
  - Make the title declarative
    - **A Study of the Effects of Chaos Theory as a Source of Complexity and Diversity in Evolutionary Processes**
    - **Chaos as a source of Complexity and Diversity in Evolution**
- General advice for abstracts (in 250 words or less)
  - What did you do ? (and why ?)
  - What methods did you use ?
  - What results have you generated ?
  - Briefly, what are you main conclusions ?

Writing a research paper

- Title
- Abstract
- Introduction - Context
- State-of-the-art
- Procedure
- Findings (the evidence)
- Conclusions
- Recommendations
- Acknowledgments
- References/Bibliography
- Appendices

Writing a paper/report

- Title page
- Acknowledgments
- Summary
- Table of Contents
- Introduction - Terms of Reference - Scope
- Procedure
- Findings (the evidence)
- Conclusions
- Recommendations
- References/Bibliography
- Appendices

Writing a paper/report

- Title
-
- Summary
-
- Introduction
- Procedure
- Findings
- Conclusions
-
- References

Writing a paper/report

- Title
- Summary or Abstract

…use the Section Headings from the lab notes

…put the requested information & figures in each Section

…compare & comment

- Conclusions
- References

Writing a paper/report

- Summary
  - most important information from the whole report
  - include some background information
- Abstract
  - a sentence for each of Objective, Technology/Method, Results, & Conclusions
  - a list of keywords
- Conclusions
  - significant findings of the work
  - note any implications
  - do not replicate the Abstract

Writing a paper/report

- Figures present data/information in a more readable form than tables of numbers

- Captions give:
  - the Figure number for identification
  - a brief explanation of the data/information in the Figure
  - a brief explanation of the significance of the data/information in the Figure
- casual reader might flip through a report just looking at the Figures…